StockSmart

Smart Inventory Management for a More Efficient Tomorrow

Our Goal

To help businesses manage their inventory efficiently, reduce stockouts and avoid overstocking.

Product Management

Add, update and manage product details (name, SKU, category, price, etc.)

Location Management

Manage multiple warehouses and store locations

Inventory Management

Track stock levels for each product at each location



Supplier Management

Maintain supplier information and contacts

Reports & Analytics

Get insights on stock levels, trends and performance

Easy to Use Interface

Simple and intuitive REST APIs for seamless integration

Secure & Reliable

Built with Spring Boot and MySQL for data security and stability

How it works

1. **Add Product** — Create product with details like name, SKU, category, price, etc.
2. **Add Location** — Register warehouse or store locations
3. **Add Supplier** — Store supplier information and contacts
4. **Manage Inventory** — Track stock for each product at each location
5. **View Reports** — Get insights and make better decisions

Key Features

Product CRUD

Add, update, delete products easily

Location CRUD

Manage warehouses and store locations

Supplier CRUD

Maintain supplier details and contacts

Inventory CRUD

Track stock quantity by product and location

Reports

View inventory data and stock status

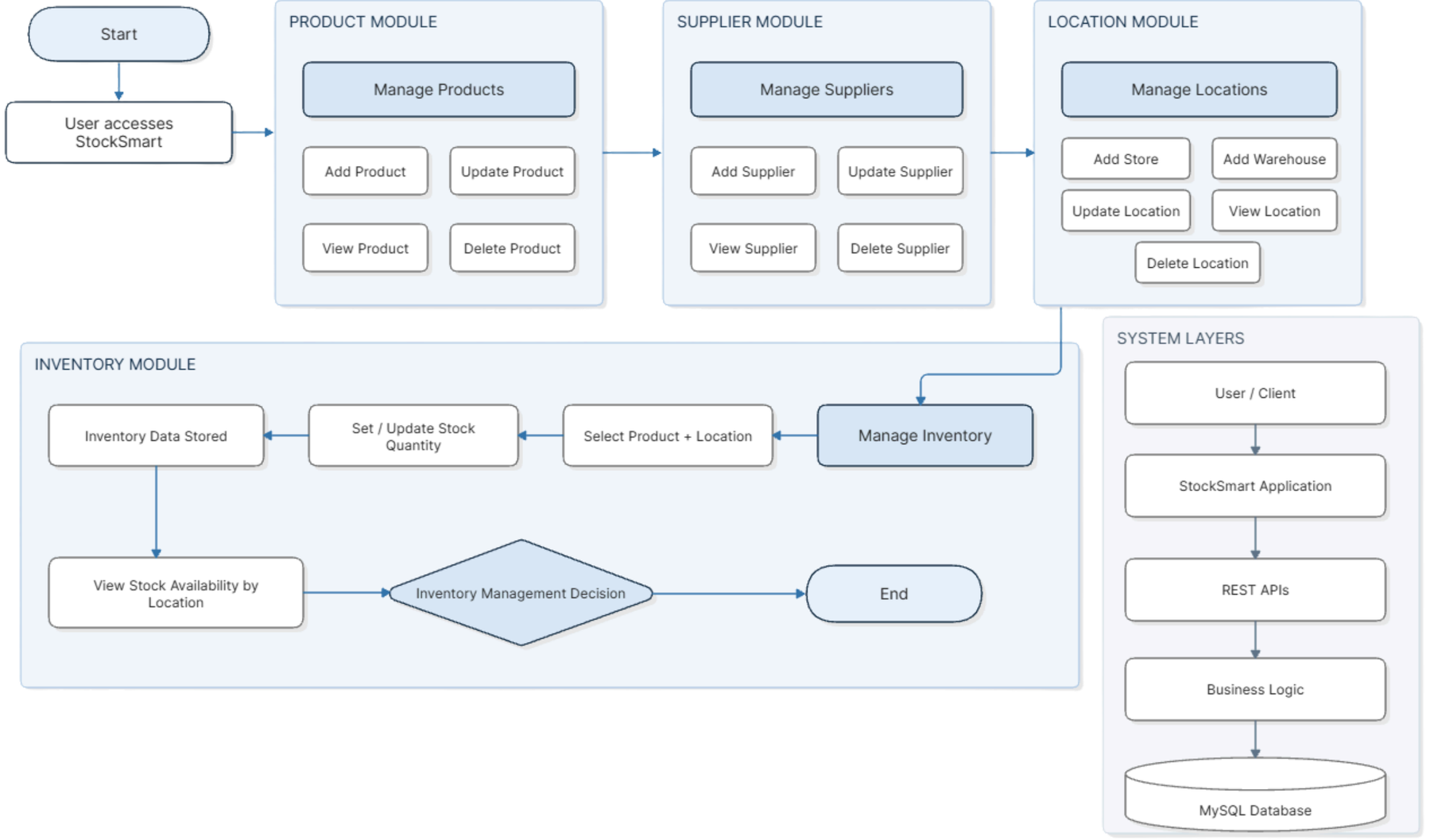
Secure APIs

Built with Spring Boot and MySQL

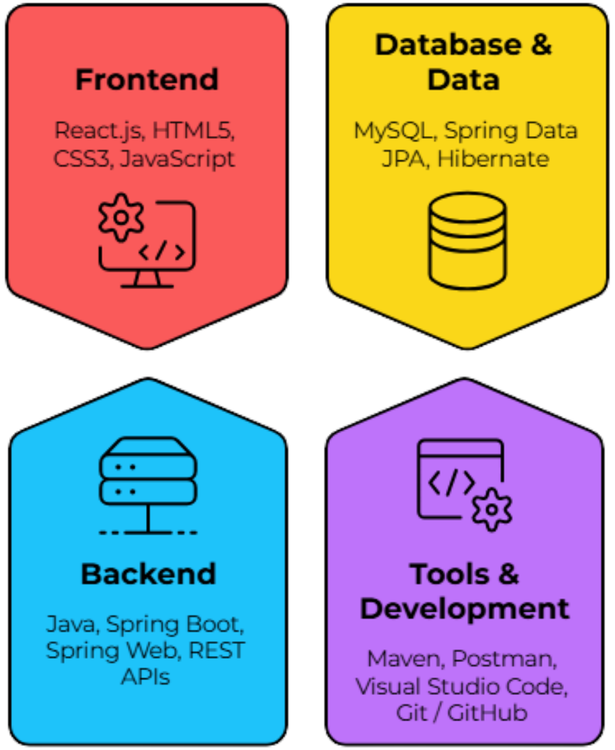
Simple Interface

Clean and easy-to-use for better management

TECHNICAL APPROACH



Tech Stack Used



FEASIBILITY AND VIABILITY

Building a Smarter and More Efficient Inventory System

Feasibility of StockSmart

The project is practical, cost-effective and can be implemented with the current resources and technology.

Technical

Built with Spring Boot, MySQL and REST APIs. Uses standard tools and well-documented frameworks.

Operational

Simple CRUD operations, easy to maintain and manage. Can be operated by a small team with basic training.

Legal

No major legal issues. Uses open-source technologies and standard development practices.

Financial

Low development and Infrastructure cost. Uses free/open-source tools and existing resources.

Schedule

Can be completed within the planned timeline with current resources and team effort.

Potential challenges and risks

Data Inconsistency

Mismatch between product, location and inventory data can lead to incorrect stock information.

Location Accuracy

Wrong or missing location details can cause stock to be recorded at the wrong place.

User Errors

Incorrect data entry or manual mistakes can lead to inaccurate inventory records.

System Downtime

Server or database issues can affect availability and operations.

Scalability

Handling large data or future expansion may require additional resources and optimization.

Strategies for overcoming these challenges

Data Validation

Use input checks, constraints and proper error handling to maintain data accuracy.

Standardized Location Mapping

Use fixed location IDs and dropdowns to avoid manual errors.

User Training

Provide clear instructions and a simple interface to reduce mistakes.

Backup & Monitoring

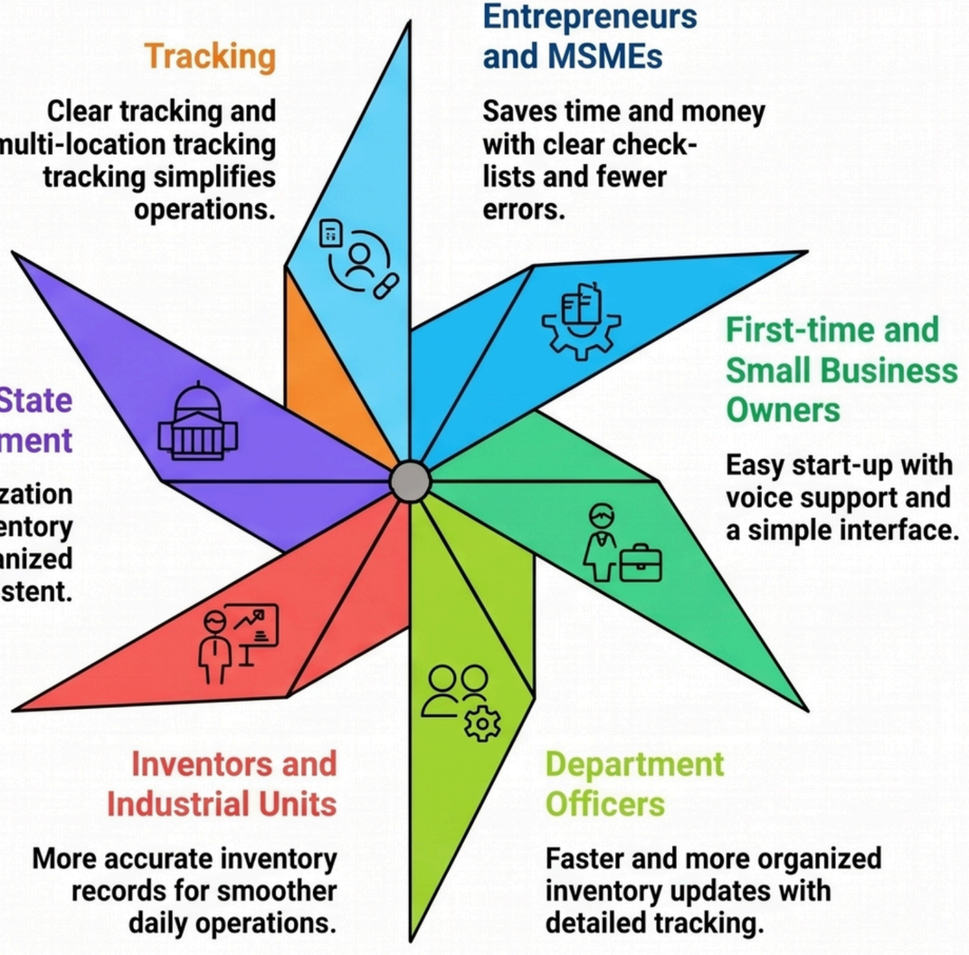
Use reliable hosting, regular backups and monitor system health for minimal downtime.

Modular Design

Design a scalable architecture to handle growth and future features.

IMPACT AND BENEFITS

Potential impact on the Target Users



Tracking

Clear tracking and multi-location tracking tracking simplifies operations.

Entrepreneurs and MSMEs

Saves time and money with clear check-lists and fewer errors.

First-time and Small Business Owners

Easy start-up with voice support and a simple interface.

State Government

Better data organization keeps inventory information organized and consistent.

Inventors and Industrial Units

More accurate inventory records for smoother daily operations.

Department Officers

Faster and more organized inventory updates with detailed tracking.

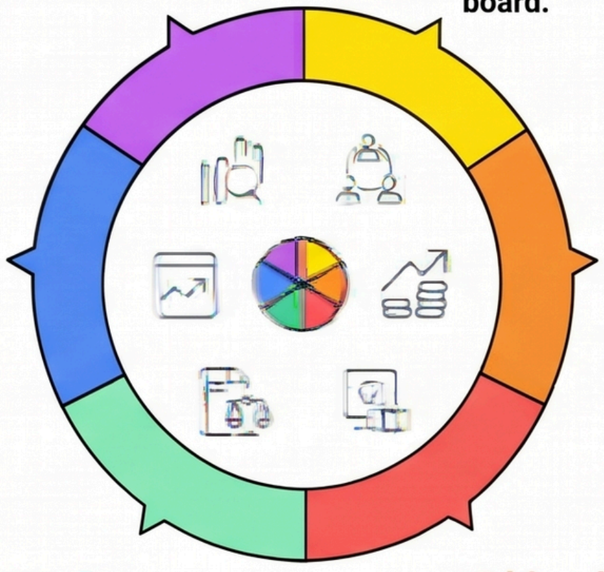
Benefits of StockSmart

Scalable Architecture

Layered Spring Boot architecture supports future expansion and growth.

Operational Control

Empowers better tock control and reducd manual tracking effort acass the board.



Future Optimization

Provides a foundation for low-stock alerts, reporting, and demand forecasting.

Multi-Location Tracking

Track product quantities separately and accurately across all different locations.

Centralized Management

Manage products, suppliers, locations, and inventory in one unified system.

Better Data Organization

Structured database keeps inventory information organized and consistent for better decision-making.

Reduced Manual Effort

REST APIs simplfy and standardize invntory operations for inreased efficiency.

RESEARCH AND REFERENCES



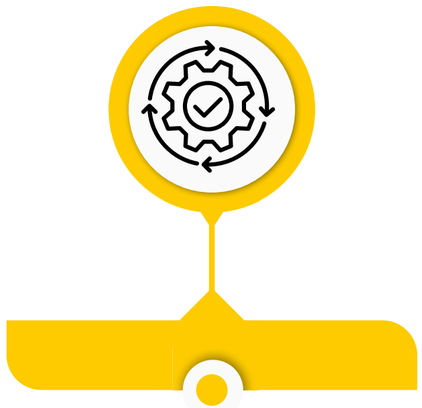
Centralized stock data provides better visibility and reduces fragmented inventory records.



Tracking inventory by location helps businesses know what stock is available and where.



Accurate and up-to-date inventory records help reduce stock discrepancies and improve stock visibility.



Digital inventory systems reduce repetitive manual tracking and make routine stock operations more efficient.



Structured inventory data provides a foundation for reporting, replenishment planning, and future demand forecasting.